GEN-3 SAFE HELICOPTER

DOCUMENT 00 — CONTROLLED BASELINE & SYSTEM ARCHITECTURE

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Program: 10-ton-class modular heavy-lift VTOL development concept

Revision: Rev A   **Year:** 2026

Status: Engineering design record

Maturity: Concept / Preliminary Design boundary

Configuration: FIXED ULTRA SAFE CAPSULE

Critical boundary: the capsule in this document is fixed to the aircraft through multiple mechanical spring/damper arms. It does **not** eject, hover independently, or use under-capsule lift fans. The previous 200-ton-class concept is not part of this baseline.

1. CONTROLLED BASELINE CONFIGURATION

Parameter	Baseline	Status
Platform	GEN-3 SAFE HELICOPTER	[T]
Development class	10-ton-class	[T]
Envelope	17 m × 17 m × 4 m nominal	[T]
Primary lift	7 distributed rotors	[T]
Rotor diameter	5.60 m nominal each	[T]
Fuel tank	1,600 L nominal	[T]
Occupant capsule	Central fixed Ultra Safe Capsule	[T]
Capsule ejection	Not included	[T]
Capsule underbody fans	None	[T]
Capsule attachment	Multiple spring/damper mechanical arms	[T]
Exact gross mass	Not closed	[AV]
Exact empty mass	Not closed	[AV]
Rotor thrust / power	Not established	[AV]
Endurance	Not established	[AV]
Certification	No certification claimed	[AV]
Physical prototype/test	None supplied	[AV]

1.1 Configuration control

This baseline is intentionally smaller and less complex than the cancelled 200-ton-class concept. Earlier Documents 00–05 for the 200-ton configuration are treated as cancelled for this program direction. Any future ejectable-capsule variant must be a separate controlled configuration, not a silent change to this document.

1.2 Data classification

[V] Verified/tested; [C] Calculated/derived; [A] Assumed; [T] Design target; [AR] Allocated requirement; [AV] Awaiting validation; [OPEN] unresolved/conflicting. No target or assumption is presented as verified.

2. SYSTEM ARCHITECTURE

The aircraft uses seven distributed primary lift rotors around a protected central occupant capsule. The architecture emphasizes modular propulsion, redundant critical functions, controlled fault isolation, crash-energy management, fire protection and emergency egress.

The capsule is a fixed occupant-survival subsystem. Multiple mechanical arms with spring/damper elements provide the intended energy-management interface between capsule and airframe. Exact stiffness, damping, stroke, peak loads, fatigue life and failure behavior require engineering closure.

The fixed-capsule configuration deliberately excludes the six 10 m capsule fans previously associated with the larger concept. There is no independent capsule flight function in Document 00.

3. PRIMARY LIFT SYSTEM

Item	Baseline	Status
Rotor count	7	[T]
Rotor diameter	5.60 m	[T]
Arrangement	Distributed / symmetric concept	[T]
Thrust per rotor	To be calculated	[AV]
Total thrust	To be calculated	[AV]
Rotor RPM	To be selected	[AV]
Tip speed	To be calculated	[AV]
Disk loading	To be calculated	[AV]
Installed shaft/electric power	To be calculated	[AV]
Transmission	Open trade	[OPEN]

Geometric calculation: one 5.60 m rotor has a disk area of approximately 24.63 m²; seven rotors provide approximately 172.42 m² of summed geometric disk area. This is only a geometric [C] result. It does not establish aerodynamic lift, useful disk area, rotor interference, power requirement, control authority or flight performance.

Required next analysis: hover induced power, profile power, rotor interference, figure of merit, forward-flight performance, transient power, one-rotor-out cases, structural loads and control authority.

4. PROPULSION SELECTION FRAMEWORK

No engine is declared selected in this baseline. A defensible engine selection requires a closed preliminary mass range and rotor power model first. Candidate architectures should include turboshaft direct/mechanical drive and turbogenerator/electric drive where technically justified.

Selection criterion	Required evaluation
Installed mass	Engine + gearbox/transmission + accessories
Continuous power	Mission hover/cruise requirement
Peak/emergency power	Takeoff, climb and degraded cases
Fuel consumption	Mission-specific SFC / fuel burn
Shaft speed	Compatibility with rotor/gearbox
Thermal management	Heat rejection and installation
Redundancy	Engine and power-distribution failure cases
Maintenance	Module replacement and inspection
Certification	Applicable engine/aircraft certification path

5. FUEL SYSTEM

Fuel capacity: 1,600 L nominal [T]. Fuel type, usable volume, tank mass, crash-resistant architecture, fuel flow, reserve and endurance remain [AV].

Crash-resistant fuel-system development is a major safety workstream. FAA rotorcraft guidance identifies crash-resistant fuel systems and occupant protection as important safety areas, including emergency-landing and fuel-system crash-resistance provisions. ■cite■turn0search12■turn0search0■

6. FIXED ULTRA SAFE CAPSULE

6.1 Architecture

The capsule is centrally located and permanently attached in this configuration. It is intended to create a protected occupant volume and a controlled structural load path. No claim is made that the capsule guarantees survival in a crash.

6.2 Mechanical isolation

Parameter	Status
Number of spring/damper arms	[AV]
Spring stiffness	[AV]
Damping coefficient	[AV]
Stroke	[AV]
Peak transmitted load	[AV]
Energy absorption	[AV]
Fatigue life	[AV]
Fail-safe behavior	[AV]
Attachment load paths	[AV]

The key engineering question is not simply whether springs exist, but whether the complete capsule-airframe-seat system provides a controllable load path under defined impact pulses. This requires dynamic FEA and physical testing.

7. TEN-LAYER SAFETY ARCHITECTURE

The ten layers below are design architecture only. They do not constitute demonstrated safety or certification.

Layer	Function	Design intent
1	Early detection	Detect fire, propulsion, electrical, thermal and structural anomalies.
2	Fault confirmation	Cross-check sensors and distinguish transient from persistent faults.
3	Fuel isolation	Isolate affected fuel paths where applicable.
4	Electrical isolation	Prevent electrical fault propagation.
5	Fire containment/suppression	Contain and suppress localized fire.

Layer	Function	Design intent
6	Propulsion degradation management	Reconfigure available propulsion/control resources.
7	Redundant critical control/power paths	Provide separated paths where safety analysis requires them.
8	Crash-energy management	Manage impact through structure, capsule interfaces, seats/restraints.
9	Emergency landing/water survival	Provide validated emergency landing, flotation and egress provisions where required.
10	Emergency egress/occupant survivability	Accessible exits, restraints, emergency equipment and evacuation provisions.

The layers must be assessed for common-cause failures. Duplication alone does not create independence. FHA, FMEA/FMECA, fault-tree analysis and system safety analysis are required.

FAA identifies crash dynamics as a dedicated technical discipline for rotorcraft and emerging VTOL aircraft and emphasizes analytical and test-based occupant-protection development. ■cite■turn0search2■turn0search7■

8. FIRE SAFETY

Conceptual sequence: detection → confirmation → fuel isolation → electrical isolation → containment/suppression → propulsion degradation/reconfiguration → emergency landing. The fixed capsule cannot separate, so its safety case must rely on hazard segregation, barriers, protected routing, suppression and crashworthiness.

No claim is made that the system is fireproof or that occupants are guaranteed to survive a fire.

9. ELECTRICAL REDUNDANCY

Critical emergency functions should use separated power and control paths. Exact generator count, batteries, buses, controllers, circuit protection and routing are [AV]. Physical segregation and common-cause analysis are required.

A future architecture should map every critical command to its power source, wiring route, controller, sensor dependencies and fallback mode.

10. CRASHWORTHINESS

The fixed capsule concept is intended to manage impact energy using the capsule structure, capsule-to-airframe spring/damper interfaces, seats/restraints and controlled structural load paths.

Verification should progress from analysis to component tests and full-scale tests as appropriate. FAA occupant-protection material describes emergency-landing and dynamic occupant-protection requirements for rotorcraft. ■cite■turn0search12■turn0search13■

11. WATER SURVIVAL AND EGRESS

If ditching/water operation remains a mission requirement, flotation, stability, water ingress, emergency exits and evacuation must be separately engineered and verified.

FAA material on rotorcraft ditching discusses flotation, trim, accessible exits and demonstration/test/analysis of egress capability under relevant conditions. ■cite■turn0search3■turn0search10■

12. PRELIMINARY MASS CLOSURE

Mass group	Status
Airframe structure	[AV]
7 rotors / hubs / blades	[AV]
Engines / powerplants	[AV]
Gearboxes / transmissions	[AV]
Fuel system / 1,600 L capacity	[AV]
Electrical generation/distribution	[AV]
Avionics / flight controls	[AV]
Fixed capsule	[AV]
Spring/damper interfaces	[AV]
Landing gear	[AV]
Fire/emergency systems	[AV]
Occupants / payload	[AV]
Margin	[AV]
TOTAL	[AV]

The 10-ton value is a development-class target, not a closed gross mass. Engine selection must wait for convergence of mass, rotor aerodynamics and power requirements.

13. REQUIREMENTS TRACEABILITY

ID	Requirement	Implementation	Verification	Status
REQ-001	7 primary rotors	7 × 5.60 m nominal rotors	Analysis/test	PENDING
REQ-002	10-ton-class target	Bottom-up mass closure	Analysis	PENDING
REQ-003	1,600 L fuel capacity	Fuel system design	Inspection/analysis	PENDING
REQ-004	Fixed protected capsule	Central capsule + spring/damper arms	Analysis/test	PENDING
REQ-005	No capsule underbody fans	Explicit exclusion	Inspection	COMPLETE
REQ-006	Ten safety layers	Safety architecture	FHA/FMEA/FTA/test	PENDING
REQ-007	Emergency egress	Exits and evacuation paths	Analysis/demo/test	PENDING
REQ-008	Crash-energy management	Structure + capsule interfaces	Analysis/test	PENDING
REQ-009	Fire hazard reduction	Detection/isolation/suppression	Analysis/test	PENDING

14. VERIFICATION & TEST PLAN

Subsystem	Progression
Rotor aerodynamics	Analysis → subscale/component test → flight test
Rotor structural loads	FEA → component/rotor testing
Powerplant integration	Power model → ground integration test
Fuel system	Hazard analysis → component/system testing
Capsule structure	FEA → component test → full-scale test
Spring/damper mounts	Dynamic + fatigue testing
Fire protection	Fire hazard analysis → system test
Electrical redundancy	FMEA/FTA → hardware-in-loop/system test
Emergency egress	Analysis → demonstration/test
Water survival, if retained	Hydrodynamic analysis → flotation/ditching tests

15. OPEN ISSUES AND RISKS

Risk	Impact	Required action	Status
Mass exceeds target	High	Bottom-up mass closure	OPEN
Power exceeds available installation	High	Rotor/power/engine trade	OPEN
Rotor interference/control complexity	High	Aerodynamic + flight-dynamics study	OPEN
Capsule interface loads too high	High	Dynamic FEA + testing	OPEN
Common-cause electrical failure	High	Architecture + FMEA/FTA	OPEN
Fire propagation to capsule	High	Fire hazard analysis + testing	OPEN
Water mission expands certification scope	Medium/High	Define mission/certification basis	OPEN

16. IP / KNOW-HOW TRACEABILITY

Candidate IP areas are recorded only as technical opportunities. No patentability, novelty or freedom-to-operate conclusion is made.

Candidate	Technical mechanism	Evidence required	Status
IP-00-01	Fixed survival capsule + multi-point spring/damper isolation	Load-path analysis, drawings, test evidence	Candidate
IP-00-02	Layered propulsion/fire/electrical containment	Architecture, FMEA/FTA, test evidence	Candidate
IP-00-03	Modular propulsion safety architecture	Module definition, interfaces, failure analysis	Candidate
IP-00-04	Integrated crash-energy + egress architecture	Occupant/structural analysis and test	Candidate

17. CONFIGURATION CHANGE CONTROL

Any future ejectable-capsule version, independent capsule fans, or other major architectural change must be issued as a separate configuration proposal with current value, proposed value, rationale, mass impact, performance impact, safety impact, certification impact, IP impact, verification requirement and approval status.

The fixed-capsule aircraft should remain a clean, simpler flagship baseline for early engineering validation.

18. ENGINEERING MATURITY

Current maturity: CONCEPT / PRELIMINARY DESIGN boundary [A]. The document is an engineering architecture and configuration record. It is not a type design, airworthiness approval, production drawing package, certified aircraft or verified performance report.

19. ENGINEERING STATUS SUMMARY

Item	Status
Document status	Concept / Preliminary Design boundary
Baseline status	Controlled for this configuration; numerical closure OPEN
Traceability	PARTIAL / PENDING
Evidence	Concept architecture; no physical prototype/test evidence supplied
Major targets	10-ton class; 7 × 5.60 m rotors; 1,600 L fuel; 17 × 17 × 4 m envelope
Major open issues	Mass, power, aerodynamics, structure, crashworthiness, fire, water, controls, certification basis
Safety status	Concept / analytical work required
Verification	Planned
IP status	Candidate / technical review

20. PRIORITIZED NEXT ACTIONS

1. Close the bottom-up mass budget and establish a realistic 10-ton target range.

2. Develop the seven-rotor aerodynamic and power model.

3. Compare turboshaft and turbogenerator/electric architectures after the power loop is established.

4. Define capsule geometry, occupant volume and all spring/damper load paths.

5. Create FHA/FMEA/FMECA/fault-tree architecture for the ten safety layers.

6. Develop the fuel and fire-protection architecture.

7. Define emergency exits, restraints and crashworthiness criteria.

8. Define water-survival mission requirements if retained.

9. Perform DFMA and modular-maintenance studies.

10. Establish Document 01 requirements and interface-control baselines.

21. FINAL QUALITY GATE

■ Fixed-capsule baseline explicitly controlled.

■ No under-capsule lift fans included.

■ No capsule ejection or independent hover capability claimed.

■ Targets and awaiting-validation data are not presented as verified.

■ No fabricated test, certification or patent status.

■ Safety architecture is clearly conceptual until verified.

■ Configuration changes are separated from the baseline.

■ Engineering maturity matches available evidence.

END OF DOCUMENT 00 — GEN-3 SAFE HELICOPTER — REV A